VALIDATION SPEAKER EXPOSURE AND CHECKPOINT SELECTION IN SPEECH EMOTION RECOGNITION

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ABSTRACT

Validation speaker separation and checkpoint selection are often discussed together, although validation optimism and common-test performance are different quantities. We compare seen- and unseen-speaker validation using cross-entropy (CE) or unweighted average recall (UAR) along identical WavLM training trajectories. A frozen protocol covers 360 main trajectories and 24 controls across three native emotion-recognition tasks. With CE selection, seen-speaker validation yields higher common-test UAR by 0.935 percentage points on CREMA-D and 2.247 on SUBESCO; the corresponding CE-versus-UAR interactions are 0.937 and 2.560 points. These four contrasts survive six-test Holm correction. RAVDESS estimates remain uncertain. Descriptive seen-minus-unseen differences under UAR selection are close to zero, without establishing equivalence. Complete 15-epoch curves and whole-group controls constrain mechanistic interpretations. The results support criterion-dependent selection effects under this fixed program, rather than a universal advantage of speaker-independent validation.

Index Terms— speech emotion recognition, speaker independence, checkpoint selection, validation, evaluation

1. INTRODUCTION

Speech emotion recognition (SER) is often evaluated on speakers not used for training. Speaker separation is an established evaluation concern. SERAB and Antoniou et al. explicitly separate training, validation and test speakers [1, 2]. More generally, optimizing a finite-sample selection criterion can affect subsequent performance evaluation [3]. However, a validation score difference and a difference in the selected models' performance on an identical unfamiliar-speaker test are distinct quantities. A larger validation–test gap alone does not establish that the corresponding selection rule produces a worse test model.

Prior SER studies already compare random and actor-independent evaluation across datasets, models or repeated runs [4, 5, 6]; matching training counts across speaker/text conditions also has precedent [7]. Such comparisons establish the relevance of evaluation protocols, but may change training or test membership along with the boundary of interest. We therefore study a specified checkpoint-selection comparison: validation on familiar versus unfamiliar speakers, each using either cross-entropy or unweighted average recall (UAR), along one shared training trajectory and on one shared speaker-exclusive outer test. Holding the candidate trajectory fixed removes differences between training runs from this within-trajectory comparison. The validation groups still contain different people, so the contrast concerns the complete selection policies rather than an isolated voiceprint mechanism.

This design follows earlier analyses of corpora already used in this project [8]. It fixes one model configuration, all 15 candidate epochs and the analysis before the new fits. The principal quantities are the common-test difference between the two cross-entropy policies and its interaction with the choice of cross-entropy versus UAR. All three native corpus tasks and all six planned tests are retained. Full trajectories and a fixed-last-epoch training-group exchange provide descriptive context; they do not supply additional hypothesis tests or determine which results are reported.

2. EXPERIMENTAL DESIGN

2.1. Shared-trajectory checkpoint-selection protocol

The protocol specifies 360 main trajectories: 24 draws and five outer folds per corpus, plus 24 paired CREMA-D control fits. Checkpoint rules share a training trajectory and speaker-independent outer test set. CREMA-D, SUBESCO, and RAVDESS retain six, seven, and eight native classes and 91, 20, and 24 speakers, respectively [9, 10, 11]. These corpora have prior research exposure. The protocol uses a private pre-execution source/plan freeze, not public preregistration. New draws do not create previously untouched data.

Within each draw, metadata hashing partitions speakers into five outer folds, stratified by recorded sex; each speaker enters exactly one outer fold. Outside each fold, disjoint development groups A and B are sex-balanced and complete across chosen texts and native classes. Eligible speakers within each recorded-sex stratum are hash-ranked using phase, corpus, draw, and fold identifiers; successive

Table 1. Prespecified per-context allocation. A and B are equal-sized, sex-balanced groups; counts denote distinct speakers or recordings.

Corpus	Classes	A/B	Outer	Texts[a]	Fit	Query[b]
CREMA-D	6	24/24	17/19	4/2	576	288
SUBESCO	7	8/8	4	4/2	224	112
RAVDESS	8	8/8	4/6	1/1	64	64

[a] Fit/query text counts. [b] Recordings in each of A and B query; outer queries contain available CREMA-D representatives, 56 SUBESCO recordings, or 32/48 RAVDESS recordings. Slash-separated outer counts denote alternative fold sizes.

equal quotas form A and B, without model scores. All query groups share texts disjoint from fit texts (Table 1); unused speakers remain excluded. Frozen representatives are MD-preferred/otherwise XX for CREMA-D, T1 for SUBESCO, and normal-intensity R1 for RAVDESS, without substitute takes. Outer queries retain available representatives, requiring all native classes for every outer speaker. Insufficient development capacity causes metadata failure, not outcome-dependent resampling.

Main trajectories train on A. WavLM base-plus [12] has a native-class linear head over mean-pooled final features; only the top four Transformer layers and head are trainable. The earlier study's fixed cfg3 uses AdamW, encoder/head learning rates $5 \times 10^{-5}/10^{-3}$, weight decay 0.01, batch size 16, and unweighted cross-entropy. All 15 epochs traverse the fit set, without early stopping or hyperparameter search. Training uses FP16 autocast with FP32 parameters; evaluation uses FP32. Mono 16 kHz audio is silence-trimmed at 30 dB; training uses three-second crops or zero-padding, and evaluation retains at most the first ten seconds.

A queries provide seen-speaker validation; B queries provide unseen-speaker validation. All 15 epochs record A, B, and outer logits. Four rules minimize mean cross-entropy or maximize UAR on A or B; exact ties select the earliest epoch, using rational recall to preserve exact UAR ties. UAR is macro recall over native classes in the whole query group, not average speaker UAR. Each group has separate sorted batches of 16, fixed across epochs and paired models. Outer audio cannot pad validation batches. Pooling lacks a length mask: padding context is fixed, but padding effects remain.

Let $T(g, q)$ be outer UAR in percent at the epoch selected by group g (seen s, unseen u) and criterion q. Per-context contrasts are

$$\delta_q = T(s, q) - T(u, q), \quad q \in \{\text{CE}, \text{UAR}\}, \tag{1}$$

$$J = \delta_{\text{CE}} - \delta_{\text{UAR}}. \tag{2}$$

These are outer UAR differences in percentage points (pp), not loss or validation-optimism ($V - T$) differences. They compare selection rules, without identifying a unique speaker mechanism. Four rules reuse one trajectory; they do not require four training runs.

Within each corpus, five fold effects are averaged equally per draw. Two-sided one-sample t inference uses 24 complete draw effects ($df = 23$). The six-test family comprises (δ_{CE}, J) in three corpora, with Holm control at 0.05 and pointwise 95% t intervals. Undefined zero-variance tests remain NA, retaining their family slots. Absolute UAR and δ_{UAR} are descriptive. The 1 pp practical reference is not an equivalence margin; no equivalence test is performed. Inference conditions on the finite corpus and training program; folds, recordings, repeated speakers, and rules are not extra replicates. Different native tasks preclude pooled inference or language-effect claims.

CREMA-D fold 0 adds one B-trained model per draw, sharing initial weights, matched stochastic schedules, and sex/text/class fit-slot budgets with A. Within each sex, slots also match speaker rank, take, and intensity. B trains for 15 epochs and records only final queries, without checkpoint selection. For whole-group query UAR Q_A, Q_B in percent,

$$\begin{aligned} E = \tfrac{1}{2}\{&[Q_A(M_{A,15}) - Q_A(M_{B,15})] \\ &+ [Q_B(M_{B,15}) - Q_B(M_{A,15})]\}. \end{aligned} \tag{3}$$

The 24 pairs and their mean are descriptive. This replaces whole training groups, not individual speakers; these queries also select main-A checkpoints.

Four technical pilots are excluded from the 384 fits. Scoring requires complete source, artifact, checkpoint, and ledger verification. Prespecified descriptions retain selected-epoch distributions and agreement, all 15 epochs' outer curves, final UAR, epochs 8–10 minus 14–15 mean UAR, and shortfall from each trajectory's maximum outer UAR. This oracle neither selects retained checkpoints nor extends training. Full processing, randomization, and verification details accompany the frozen protocol.

3. RESULTS

3.1. Prespecified common-test effects

Table 2 reports the complete six-test family. Seen-CE selection exceeds unseen-CE selection by 0.935 pp on CREMA-D and 2.247 pp on SUBESCO; their pointwise intervals exclude zero and Holm-adjusted p values are 0.0264 and 0.0083. Positive interactions of 0.937 and 2.560 pp also survive Holm correction. Thus the seen–unseen selection contrast depends on the criterion in these two tasks. CREMA-D's CE estimate is below the 1 pp reference; SUBESCO's exceeds it, but both intervals include 1 pp. These tests do not establish effects larger than that reference.

RAVDESS gives $\delta_{\text{CE}} = 0.486$ pp and $J = 0.469$ pp, with both intervals crossing zero. This is inconclusive evidence, not a third confirmed effect or equivalence. No corpus was dropped and no additional fits or tests were added after inspection.

Table 2. The six prespecified common-test effects in UAR pp, pointwise 95% draw-level t intervals, and two-sided raw and six-family Holm p values.

Corpus	Contrast	Mean	Pointwise 95% CI	Raw p	Holm p
CREMA-D	δ_{CE}	0.935	[0.287, 1.583]	0.0066	0.0264
CREMA-D	J	0.937	[0.210, 1.663]	0.0138	0.0413
SUBESCO	δ_{CE}	2.247	[0.945, 3.549]	0.0016	0.0083
SUBESCO	J	2.560	[1.103, 4.016]	0.0014	0.0083
RAVDESS	δ_{CE}	0.486	[-0.198, 1.170]	0.1549	0.3098
RAVDESS	J	0.469	[-0.633, 1.570]	0.3877	0.3877

Table 3. Descriptive outer UAR (%) of the four checkpoint policies and fixed epoch 15, plus δ_{UAR} in pp; five folds then 24 draws are equally weighted.

Corpus	Seen CE	Unseen CE	Seen UAR	Unseen UAR	Last 15	δ_{UAR}
CREMA-D	56.253	55.318	57.154	57.155	53.715	-0.002
SUBESCO	47.113	44.866	47.426	47.738	47.292	-0.313
RAVDESS	33.941	33.455	32.943	32.925	32.821	0.017

3.2. Absolute performance and selection behavior

Table 3 retains all four policies and epoch 15. Under UAR selection, the descriptive seen–unseen differences are −0.002, −0.313 and +0.017 pp. Their small means do not establish equivalence. UAR-selected models have higher mean outer UAR than CE-selected models in CREMA-D and SUBESCO, whereas RAVDESS shows the opposite ordering. These within-task descriptive rankings do not support a universally best criterion.

Mean seen/unseen CE epochs are 6.71/5.89, 10.71/7.92 and 14.35/13.86 in corpus order; UAR means are 10.53/10.09, 12.54/11.98 and 13.66/13.48. CE epoch agreement is 58.3%, 26.7% and 76.7%; UAR agreement is 37.5%, 27.5% and 42.5%. Similar aggregate UAR contrasts therefore need not mean selection of the same epoch. Full distributions and oracle shortfalls accompany the artifact. On CREMA-D, the mean oracle shortfall is 2.34/3.27 pp for seen/unseen CE, about 1.44 pp for either UAR rule, and 4.88 pp at epoch 15; these are descriptive shortfalls from trajectory-wise outer-test maxima, not attainable selection guarantees.

3.3. Complete trajectories and whole-group exchange

Figure 1 shows a CREMA-D plateau and continued mean outer improvement on SUBESCO and RAVDESS. The prespecified mean UAR at epochs 8–10 minus epochs 14–15 is +0.208, −2.894 and −8.977 pp. The curves therefore do not support a uniform late-training collapse explanation. RAVDESS selects epoch 15 in 72/120 seen-CE and 62/120 unseen-CE trajectories; the fixed horizon limits interpretation of later behavior.

A post hoc reconstruction restricting candidates to epochs 1–8, 1–10, 1–12, or 1–15 shows window-sensitive contrasts [8]; it adds no tests and cannot establish behavior beyond epoch 15.

The CREMA-D training-group exchange has mean E = 2.828 pp across 24 pairs. All pair values, including the negative pair, are retained in the artifact. This is consistent with an average advantage on queries from the model's own training-speaker group under matched slots. It neither isolates individual speaker causality nor establishes why CE and UAR choose different checkpoints.

4. DISCUSSION

The main finding is criterion-dependent common-test selection behavior. Speaker-exclusive validation remains an established way to evaluate generalization, but that rationale does not guarantee that its CE-selected checkpoint outperforms a seen-CE checkpoint on the same outer test. Conversely, these results do not justify replacing speaker-independent reporting with seen-speaker scores: model selection and reported evaluation answer different questions. CE measures probability quality whereas UAR depends on class decisions; their distinction offers a plausible explanation to investigate, not a tested calibration mechanism.

Earlier project studies [8] help separate the estimands. N14R2 found a 1.419 pp difference in validation optimism, $\Delta(V - T)$, while inner procedures also changed fitting membership. DUAL held fitting trajectories common and found a 2.706 pp optimism contrast while the selected outer-score difference was positive. Its post hoc fixed-configuration, final-epoch contrast was already 2.839 pp (selected minus fixed: −0.133 pp), so the optimism contrast cannot be attributed entirely to selection. These quantities are not the present δ_q. Subsequent checkpoint diagnostics motivated this fixed-configuration study; they are post hoc uses of existing predictions, not independent replication. Shared corpora and

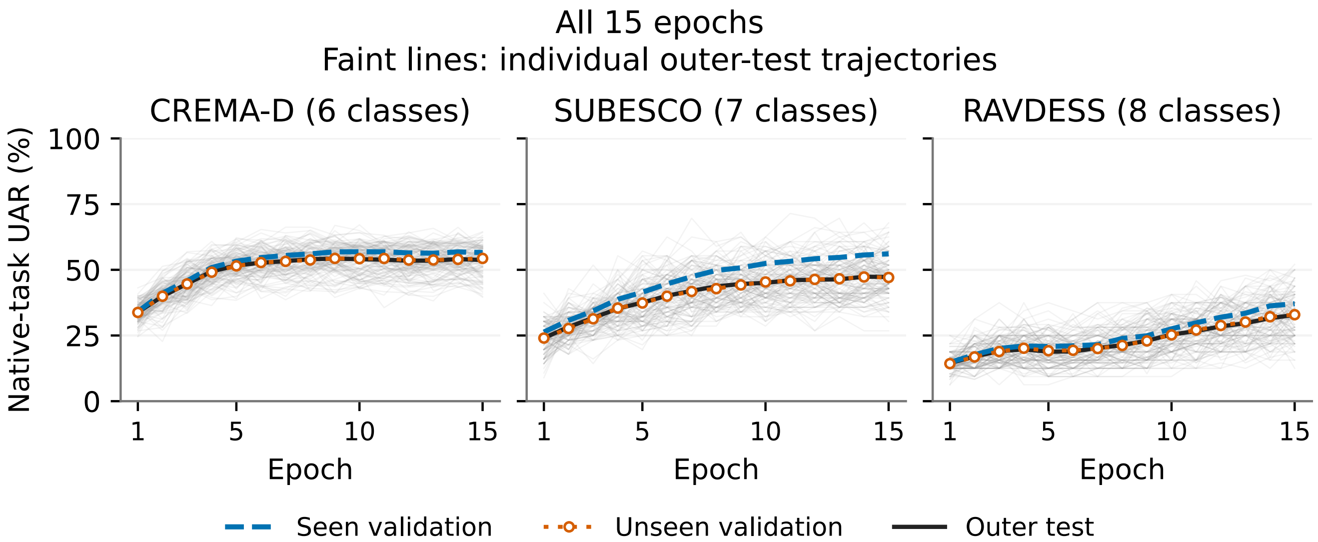

Fig. 1. All 15 epochs in the three native tasks. Faint grey lines are the 120 individual outer-test trajectories per corpus. The three heavy curves equally average the 120 main contexts for seen validation, unseen validation and outer testing. No smoothing or curve uncertainty bands are added.

previous fits do not enlarge the current inferential sample.

4.1. Scope and data-construction implications

The estimates condition on finite acted-speech corpora, one partially fine-tuned encoder, and the fixed training horizon. Tasks differ in native classes, speaker counts, texts and fit sizes; they identify neither a language effect nor a fair ranking of corpus difficulty. Validation identities and recording distributions also differ, precluding an isolated voiceprint interpretation. Fixed role batches preserve padding context without eliminating its effects. The group-exchange queries also participate in main-model selection and are not globally unused blind tests.

Separate recording-budget and coverage studies, and the synthetic-speech protocol that did not reach formal training, remain in the artifact [8]; they provide no pooled confirmation or demonstrated elimination of the seen–unseen gap.

4.2. Execution and availability

All 384 formal fits completed on one RTX 5070, with no training retries; four pilots were excluded. Unit wall times total 4.78 hours, including prescribed evaluation and saving. The artifact records 110,160 update attempts and 44 AMP skips. Full source/artifact/checkpoint/ledger verification and independent numerical replay passed. Fresh-process restoration of the four predetermined units reproduced 30 saved-state/query-group comparisons exactly, after correcting a documented launcher environment error. This limited check is not independent retraining. A portable prediction package passed numerical replay after extraction; it excludes audio, base weights and checkpoints. The artifact is access-restricted at preparation. Its private freeze establishes version history, not public preregistration.

5. CONCLUSION

On shared WavLM trajectories and outer tests, CE-based seen-speaker selection performed better on CREMA-D and SUBESCO, with evidence that the exposure contrast changes under UAR selection. RAVDESS remains uncertain, and full curves do not support uniform late-training collapse in these runs. The contribution is a controlled selection-policy comparison with criterion dependence, not a universal validation prescription or an identified speaker/language mechanism.

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